

EU LIFE WADER (LIFE20 NAT/UK/000277) Harehope Sediment Monitoring

An assessment of sediment chemical and physical properties prior to the restoration of the River Breamish

Author name(s): Justyna P. Olszewska, Gloria Dos Santos Pereira, Patrick Keenan, Rebecca McKenzie, Alice MacSporran and David Nicholls

Client: Environment Agency
Issue number 0.1
11.04.2025

UK Centre for Ecology & Hydrology

Contents

1. Introduction 3

1.1 Background 3
1.2 Aims 3

2. Methodology 4

2.1 Study design and sample collection 4
2.2 Analyses of sediment chemistry 5
2.3 Analysis of sediment physical properties 8

3. Results and Discussion 9

3.1 Sediment chemistry 9
3.2 Sediment physical properties 16

4. References 20

1. Introduction

1.1 Background

The Harehope Restoration project aims to deliver a full-scale restoration on the River Till at Harehope to improve water quality, specifically in relation to P, N and fine sediment. The project is part of the EU LIFE WADER (LIFE20 NAT/UK/000277) cross-border project, which is located mostly in Northumberland in England, but also includes the Tweed River catchment in Scotland. It aims to tackle key issues negatively affecting freshwater, dune and marine habitats within these areas

The River Till is a major tributary of the River Tweed, draining much of north Northumberland and a small area of the Scottish Borders region. In the upper catchment, including the restoration area, the River Till is known as the River Breamish. The River Till and its tributaries have been designated as the nationally important River Tweed Catchment Rivers (England: Till Catchment) Site of Special Scientific Interest (SSSI) and are also within the River Tweed Special Area of Conservation (SAC). Due to the long history of modifications, including the strengthening of the river channel in the late 18th/ early 19th century, and changes to the floodplain use, none of the seven Till SSSI units are currently in 'favourable condition'.

The project restoration measures are implemented within an approximately 3 km^2 area and include a 2 km straightened reach of the Breamish between New Bewick Bridge and Bewick Bridge.

1.2 Aims

The main aim of this project was to collect data on chemical and physical properties of sediment in the River Breamish prior to the implementation of upcoming restoration measures, which will form a baseline for an assessment of sediment responses following the restoration.

The specific aims of the project were as follows:

1) Carry out analysis of sediment physical properties
2) Analyse sediment samples for total organic carbon (TOC), total carbon (TC), total nitrogen (TN), total phosphorus (TP) and metals
3) Determine available N and available P fractions in sediment samples
4) Produce a brief report documenting the data and detailing the findings.

2. Methodology

2.1 Study design and sample collection

Sediment samples were collected by the Tweed Forum staff, who installed a single sediment trap at three river sites (Table 2.1): downstream of the restoration area (Site 1), within the restoration area (Site 2) and upstream of the restoration area (Site 3).

Samples were collected on 20th February, 19th March, 4th June, 6th August and 22nd October 2024. The original sampling plan included monthly sampling across the period of 12 month, however, this was revised following the issues raised during the first two sampling rounds, as summarised below:

- The February and March sampling showed difficulties with obtaining the sufficient amount of sediment material for all the analysis types, and with removing water from the samples. In particular, this was an issue at Site 3, which returned a small sample size in February and even smaller in March.
- Concerns were raised by UKCEH regarding the amount of the collected sediment, which was insufficient to carry out all the planned sediment analyses.
- Following the feedback from the Tweed Forum and UKCEH teams, it was decided to increase the timescale of the sample collection to 2-3 months to improve amounts of sediment accumulated in the sediment traps. In additon, the design of the sampling device was amended to decrease the likelihood of becoming blocked. This improved the amount of the collected material, however, the issue with high water content remained, and the amount of sediment in samples was lower than the sample sizes requested by UKCEH, and insufficient to carry out the full set of analyses.

Each sediment sample collected by Tweed Forum was dived into four subsamples, as required for the range of the sediment analyses, and transferred into polythene zip lock bags. The February and March samples were stored in cold conditions (5°C) prior to being transported to UKCEH within one day of the sample collection. The samples collected on the remaining occasions were stored frozen by Tweed Forum and delivered to UKCEH on 29th October 2024, following the last sampling occasion. Once delivered to UKCEH, the samples were distrubuted to the appropriate laboratories and stored frozen at -20°C prior to the analyses.

Table 2.1 Locations of the sampling sites for the baseline sediment monitoring study.

Sampling site	Site Code	NGR
1	Downstream TIMS	NU 05764 22429
2	Mid-reach TIMS	NU 05734 21755
3	Upstream TIMS	NU 07234 2072

2.2 Analyses of sediment chemistry

2.2.1 Organic carbon, total carbon and total nitrogen

Sediment samples were air-dried and pulverised in a planetary ball mill (Fritsch Pulverisette 5) to produce a homogenised sample. For those samples to be analysed for organic carbon, prior to analysis, the samples were repeatedly acidified with 1-2 ml of 3M HCl and dried until effervescence, caused by the reaction with inorganic carbon, no longer occurred. Excess acid was then evaporated off in a fume cupboard.

All samples were then oven dried at 105°C (± 5°C) for a minimum of 3 hours, then cooled and sealed prior to weighing. Each sample was weighed into a tin cup (typically 15 mg for soil) on a 6-place micro-balance and analysed for total organic carbon (TOC), total carbon (TC) and total nitrogen (TN) using a ThermoFisher Scientific FlashSmart elemental analyser. This analysis is based on a modified Dumas method, working on the principle of oxidative combustion, using separation of analytes by a GC (gas chromatography) column, followed by thermal conductivity detection. All calculations were carried out by the instrument software and results expressed in% notation.

2.2.2 Total phosphorus

Total Phosphorus (TP) was measured in the sediment samples using a H_2O_2/H_2SO_4 digestion followed by colorimetric analysis, carried out on a Seal Analytical discrete analyser.

0.36 g of air dried and ground (to 2 mm) sediment sample was weighed into boiling tubes and 4.4 mL of digestion matrix added. The digestion matrix contained sulphuric acid, hydrogen peroxide, selenium powder and lithium sulphate. This was repeated for all samples including reference samples and blanks. The boiling tubes were then placed in a block digester and the temperature was stepped to 250 ^oC and held for 15 minutes to allow the mixture to stabilise, then further stepped to 400 ^oC where the temperature was maintained for 2 hours to complete the digestion.

Once digested the samples and references were allowed to cool, then diluted to a total volume of 50 mL using ultrapure water, and mixed. After allowing the silica

residue to settle overnight, the supernatant was collected and diluted a further 5 times, and then measured colourimetrically using a SEAL AQ2 discrete analyser. Phosphorus concentrations were determined by reaction with acidic molybdate in the presence of antimony to form an antimony-phospho-molybdate complex. Ascorbic acid reduces this to the intensely blue phosphomolybdenum complex, measured spectrophotometrically at 880 nm.

Calibrations were run using standards, prepared from blank matrix digested in the same manner as the samples, using the range 0 – 2 mg L^{-1}, with control standards analysed every ten samples to check precision and reproducibility throughout the run. Blanks and reference samples to assess the chemistry and the digest were run with each batch. Data was corrected for moisture content and reported in mg kg^{-1} P.

2.2.3 Total metals and sulphur

Samples were air-dried and pulverised in a planetary ball mill (Fritsch Pulverisette 5) to produce a homogenised sample. Half a gram of each sample, including Certified Reference Materials (CRMs), were weighed individually into MARS Xpress Teflon vessels (a CEM MARS-5 microwave). The samples (including two blank solutions) were digested by a reverse aqua regia mixture of nitric acid (Fisher trace metal grade, 67-69%) and hydrochloric acid (J.T. Baker hydrochloric acid for trace analysis, 38%) in a microwave digestion system. After cooling, the samples were filtered using Whatman ashless, Grade 40 filters and made up to 50 mL with ultra-pure water.

The samples were then analysed using Inductively Coupled Plasma Mass spectrometry (ICPMS; Agilent 7900). Five internal standards (Sc, Ge, Y, In and Bi, VWR) were used to correct for sample-specific matrix effects and instrumental drift. The ICPMS analysis was performed at a x5 dilution, using matrix-matched calibration standards for 11 elements (Mn, Fe, Ni, Cu, Zn, Cd, Pb, Mg, S, V & Ca). Every sample, CRM, blank and calibration solution was analysed using three replicates.

2.2.4 Phosphorus fraction determination

The P extraction method followed Hupfer et al. (1995), based on Psenner et al. (1988), with a modification which was used by Lewandowski et al. (2003) and Hupfer et al. (2009). The forms of sediment P associated with this extraction procedure are summarised (Table 2.2).

The samples were subjected to the following sequential extraction procedure: (i) ‘labile P’ - extraction in 1 M NH4Cl for 30 min [TSP], (ii) ‘reductant-soluble P’ - extraction with 0.11 M NaHCO3 / 0.11 M Na2S2O4 for 1 h, repeated for 5 min [TSP], (iiia) extraction in 1 M NaOH for 16 h, repeated for 5 min [SRP], (iiib) supernatant from (iiia) analysed for TSP to determine ‘organic P’ [TSP] from ‘metal-oxide adsorbed P’, (iv) ‘apatite bound P’ - extraction with 0.5 M HCl for 16 h, repeated for 5 min [TSP], and (v) ‘residual P’ - digestion with 30% (v/v) H2SO4 and 8% K2S2O4 at 121 ºC for 30 min [TSP].

The sediment slurries and extraction mediums were continually shaken (60 rpm on 360º rotator; PTR-60, Grant-bio, Shepreth, UK) in the dark at 25 ºC for the periods

outlined above. Supernatants were collected following centrifugation and filtration (Whatman® GF/F filter, nominal pore size 0.7 µm) at the end of each extraction step. SRP and TSP analyses were conducted on filtered water samples according to Murphy and Riley (1962) and Eisenreich et al. (1975), respectively, using a SEAL AQ2 analyser. Total sediment P was estimated as the sum of all P fractions and percentage contribution of each sediment pool to the sediment total P was calculated.

The results of the analysis of metal adsorbed and organic fractions in the sediment collected in March at Site 1 indicated an analytical error and therefore, were omitted from this report. No sediment material was available to analyse a replicate sample.

Table 2.2. Overview of common operational sediment phosphorus (P) fractions based on sequential sediment P extraction procedures (Meis et al., 2012).

Operational fraction	Expected P species in fraction	Driver of P release	Likelihood of release under natural conditions
'*Labile P*'	Directly available P; pore water P; loosely bound or adsorbed P	Desorption; diffusion;	High
'*Reductant-soluble P*'	P bound to Fe-hydroxides and Mn-compounds	Anoxia, redox potential	High
'*Organic P*'	Allochthonous/ autochthonous material; detritus	Mineralisation (temperature)	Medium - High
'*Metal-oxide adsorbed P*'	P adsorbed to metal oxides (mainly Al, Fe); P exchangeable against OH^-	High pH (e.g. photosynthetic activity)	Medium - High
'*Apatite bound P*'	P bound to carbonates and apatite P	Low pH	Medium
'Residual P'	Refractory compounds		Low

2.2.5 Available nitrogen

The analyses of the inorganic nitrogen species (nitrate/nitrite (NO_3^-/NO_2^-) and ammonium (NH_4^+)) content in the sediment was preceded by extracting the components using the potassium chloride (KCl) method. A known volume of 1M of KCl was added to a known weight of wet sediment and shaken with a Stuart SSL1 orbital shaker for 60 minutes at a speed of 150 rpm, along with 3 procedural blanks.

Samples were then gravity filtered through 150 mm diameter filter paper, Whatman 40. Once filtered, extracts were shaken to homogenise the extract before pouring into sample bottles and storing frozen at -20°C. Extracts were defrosted shortly before analysis and analysed for NO_3^-/NO_2^- and NH_4^+ using a SEAL AQ2 analyser. Blanks and standards were run with every set of samples to check for accuracy, but it needs to be noted that the small sample size available for the extraction process introduced a certain level on uncertainty to the obtain results of the analyses.

Dry weight and soil moisture content were determined using the gravimetric soil moisture method, where sediment is weighed and then dried in an oven at 40°C until the dried sample is at a constant weight.

The soil N concentration was determined from the calculation:

$$Cs = (Ce * V) / W$$

where:

Cs = sediment N concentration (µg N g^{-1} sediment)

Ce = concentration of the extracts as determined by the SEAL AQ2 analyser (mg N L^{-1})

V = volume of KCl used + volume of water in soil sample (wet weight*soil moisture content) (mL)

W = dry mass of soil (wet weight - volume of water in soil sample) (g)

2.3 Analysis of sediment physical properties

2.3.1 Particle size distribution

The sediment samples were rinsed from their bags using 18 MΩ/cm ultrapure water (Thermo Scientific GenPure Pro UV) and placed into sterile falcon tubes. Excess water was removed via evaporation at room temperature until the samples had a paste-like consistency. Samples were then mixed thoroughly with a spatula before a representative subsample was taken and 5 mL of 5% sodium hexametaphosphate was added to each sample. Subsequently, the samples were topped up to 60 mL with ultrapure water, and each sample was vortex mixed for 30 seconds and left for 24 hours.

Prior to the analysis, each sample was vigorously shaken, and ultrapure water was used to ensure all sediment was removed from the falcon tubes. The samples were filtered through a 3mm sieve (to remove large particles) before being added to the dispersion reservoir. The samples were then analysed for the particle size distribution on a Malvern Panalytical Mastersizer 3000 instrument, using the hydro G dispersion unit.

3. Results and Discussion

3.1 Sediment chemistry

3.1.1 Carbon, nitrogen and phosphorus content

The content of total carbon (TC), total organic carbon (TOC), total nitrogen (TN) and total phosphorus (TP) in the sediment at three River Breamish sites in 2024 are shown in Table 3.1.

The lowest and highest TC was measured at Site 1 – 0.5% and 6.5%, respectively. The mean content of TC varied from 2.3% of sediment mass at Site 3 to 3.8% at Site 1. TOC accounted for most of the carbon content in sediment (87% of TC at Sites 1 and 3, and 84% of TC at Site 2).

TN content varied from 0.1% at Site 1 in October and Site 3 in February to 0.6% at Site 1 in August 2024. There was little variation in the mean TN content measured in sediment at the three sites (0.2% at Site 3 and 0.3% at Sites 1 and 2).

The TP concentrations ranged from 816 µg P g^{-1} d.w. to 1447 µg P g^{-1} d.w., with both values recorded at Site 1 in June and August 2024, respectively. The average TP concentrations across the three sampling dates varied from 979 µg P g^{-1} d.w. at Site 3 to 1131 µg P g^{-1} d.w. at Site 1.

Table 3.1 Content of Total Carbon (TC), Total Organic Carbon (TOC), Total Nitrogen (TN) and Total Phosphorus (TP) in sediment.

Date	Site	TC (% mass)	TOC (% mass)	%TOC in TC	TN (% mass)	TP (µg P g^{-1} d.w.)
20/02/2024	1	3.7	3.2	87.2	0.3	NA
	2	2.1	1.8	85.3	0.2	NA
	3	1.6	1.4	87.0	0.1	NA
19/03/2024	1	5.1	4.4	87.0	0.4	NA
	2	4.7	3.9	82.2	0.4	NA
	3	NA	NA	NA	NA	NA
04/06/2024	1	3.5	3.0	85.8	0.3	816
	2	2.9	2.4	82.4	0.2	984
	3	2.5	2.2	86.8	0.2	950
06/08/2024	1	6.5	5.4	83.0	0.6	1447
	2	5.5	4.7	83.9	0.5	1027
	3	NA	NA	NA	NA	NA
22/10/2024	1	0.5	0.4	75.5	0.1	NA
	2	3.5	2.9	83.3	0.3	1138
	3	2.9	2.3	79.3	0.2	1008
Annual mean	1	3.8	3.3	86.8	0.3	1131
	2	3.7	3.1	83.8	0.3	1050
	3	2.3	2.0	87.0	0.2	979

3.1.2 Total metals and sulphur

Table 3.2 shows concentrations of total metals and sulphur in the sediment. The mineral composition of the sediment, in particular the content of elements that have the capacity to bind phosphorus (P), such as Mn, Fe and Ca, is an important factor, as it affects the sediment P sorption capacity (Søndergaard et al, 1996). Mn and Fe oxyhydroxides can bind dissolved P under oxic conditions and release it to the pore and bottom water when conditions become anoxic (Spears et al., 2007).

The mean concentrations of manganese (Mn) at the three sites varied from 850 µg g-1 d.w. (at Site 3) to 1879 µg g^{-1} d.w. (at Site 1), with the Mn content at Site 3 being twice as low as at the remaining two sites.

The mean concentrations of iron (Fe) and calcium (Ca) showed a lower variation compared to Mn between the three sites. The mean content of Fe ranged between 24,708 µg g^{-1} d.w. at Site 3 to 28,315 µg g^{-1} d.w. at Site 2. The mean concentrations of Ca varied between 4,672 µg g^{-1} d.w to 5,686 µg g^{-1} d.w at Sites 3 and 2, respectively.

Similarly to Fe and Ca, the remaining elements displayed lower variation in their mean concentrations between the three sites than Mn, except for sulphur, the content of which at Site 3 was 1.8 and 2.1 times lower than at Sites 1 and 2, respectively.

The concentrations of copper (Cu), zinc (Zn), cadmium (Cd) and lead (Pb) were compared against the standards developed by the Canadian Council of Ministers of the Environment (Interim Sediment Quality Guidelines (ISQGs) based on Threshold Effect Levels (TELs); CCME, 2001).

The concentrations of Zn exceeded the ISQG of 123 µg g^{-1} d.w, above which adverse effects to freshwater organisms can occasionally occur, at Site 1 in June and at Sites 1 and 2 in August, with the mean Zn content at Site 1 being above the standard.

In addition, the concentration of Cd measured at Site 1 in August was above the ISQG of 0.6 µg g^{-1} d.w., indicating potential risk to organisms. The individual and mean concentrations of Cu and Pb in sediment remained below the ISQGs of 35.7 µg g^{-1} d.w and 35.0 µg g^{-1} d.w, respectively, developed for these elements.

Table 3.2. Concentrations of total metals and sulphur (µg g^{-1} d.w.) in sediment. The content of Cu, Zn, Cd and Pb is compared against Canadian Interim Sediment Quality Guidelines (ISQGs/TELs) for freshwater sediment above which adverse effects occasionally occur. The concentrations of metals which exceeded the ISQGs are written in a red font.

Date	Site	Mn	Fe	Ni	Cu	Zn	Cd	Pb	Mg	S	V	Ca
04/06/2024	1	1080	29076	41.1	25.4	131	0.34	36.3	8471	581	49.5	4654
	2	467	23443	31.7	22.6	95.9	0.29	31.1	7191	633	39.0	4287
	3	682	22385	29.3	18.0	103	0.19	29.6	6597	<LOD	38.8	3364
06/08/2024	1	4268	32639	48.3	34.8	180	0.68	41.0	9351	1130	50.0	7742
	2	3713	32695	42.4	27.1	159	0.58	39.1	8785	740	47.6	6741
	3	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
22/10/2024	1	287	13841	19.3	9.0	58.7	0.10	15.0	4118	<LOD	14.8	2004
	2	1348	28808	36.7	23.4	108	0.34	35.8	8776	628	43.8	6029
	3	1019	27229	35.4	17.0	102	0.34	24.9	8856	638	40.7	5980
Mean concentrations	1	1879	25186	36.3	23.0	124	0.38	30.8	7313	570	38.1	4800
	2	1843	28315	36.9	24.4	121	0.41	35.4	8251	667	43.5	5686
	3	850	24807	32.4	17.5	102	0.26	27.3	7726	319	39.8	4672
ISQGs/TELS					*35.7*	*123*	*0.60*	*35.0*				

3.1.3 Phosphorus composition

Figure 3.1, Figure 3.2 and Figure 3.3 present the contributions of each sediment pool expressed as a percentage of total P. The concentrations of P fractions in the sediments at the three sites are shown in Table 3.3. These data provide a useful information on the P reactivity and potential bioavailability in aquatic ecosystems (Kozyrev et al., 2023), as P compounds differ in their likelihood of release from sediment into water under natural conditions and the drivers that control this process, as listed in the 2.2, Section 2. Apatite bound P was the dominant P pool in the sediment at all three sites, accounting on average for 45%, 70% and 63% of total phosphorus (TP) at Sites 1, 2 and 3, respectively. This fraction of P is more stable and has lower likelihood of release under natural conditions compared to labile P, organic P or P bound to metals.

At Site 1, the reductant soluble P, which represents a dominant release sensitive pool including Fe- and Mn- bound P complexes, had the second largest contribution to P content, accounting on average for 21% of TP, with the P bound to metal oxides contributing 17% to TP. Proportions of the reductant soluble P pool and the metal adsorbed pool at Site 2 were similar – 11% and 10%, respectively – whilst at Site 3, these two P pools contributed on average 13% to the TP content. The highest mean content of organic P pool was measured at Site 1 (15%), compared to 7% at Site 2 and 9% at Site 3. The labile and residual P pools accounted for less than 1% and 2% of the TP content, respectively, at each site.

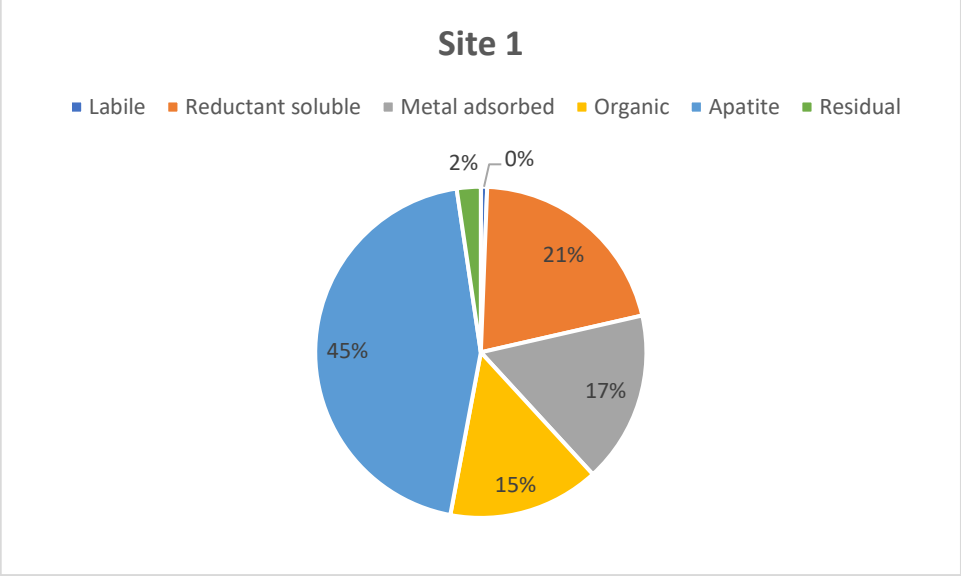


Figure 3.1 Contributions of the sediment pools expressed as a percentage of total P at Site 1.

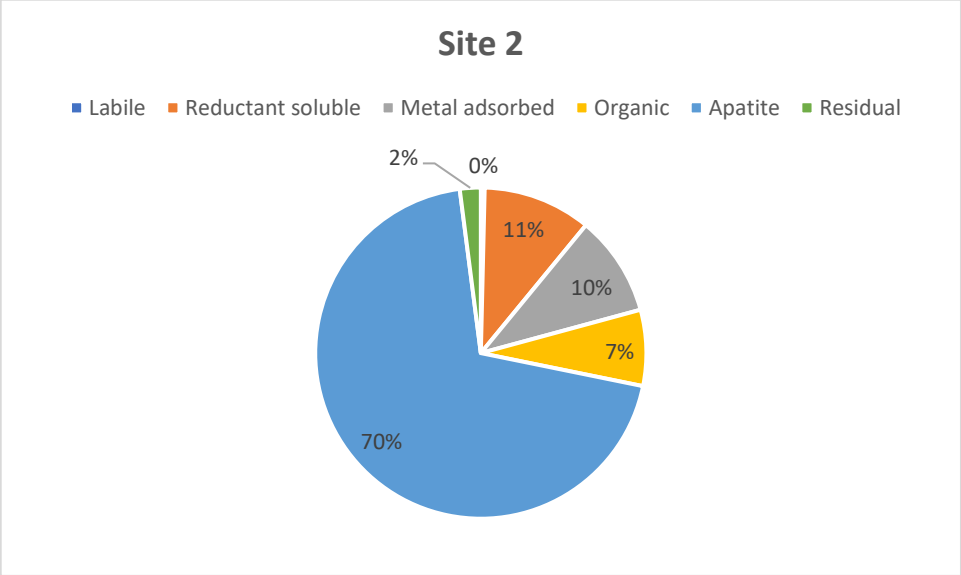


Figure 3.2 Contributions of the sediment pools expressed as a percentage of total P at Site 2.

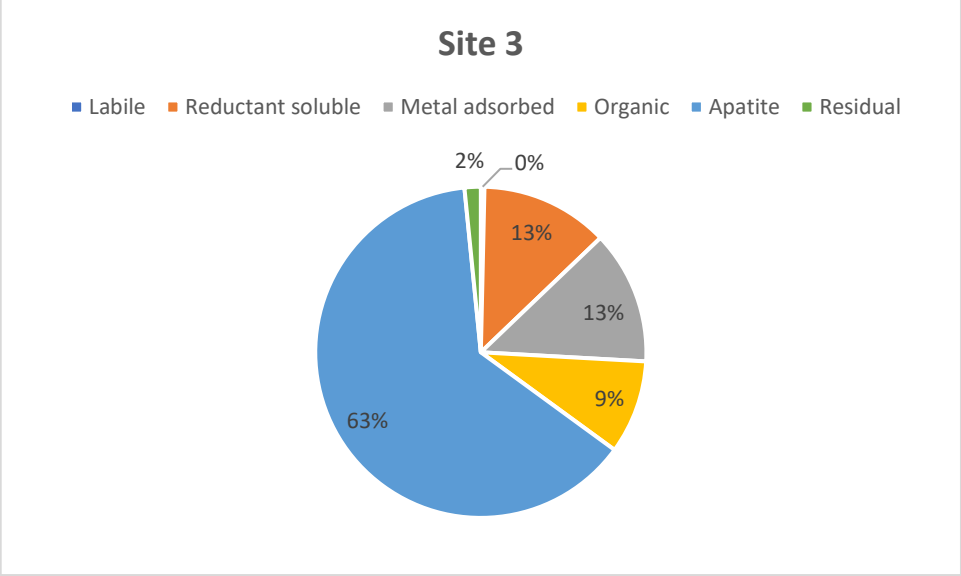


Figure 3.3 Contributions of the sediment pools expressed as a percentage of total P at Site 3.

Table 3.3 Concentrations of the phosphorus pools in the sediment.

Date	Site	Labile	Reductant soluble	Metal adsorbed	Organic	Apatite	Residual
		(µg P g^{-1} dry weight sediment)					
20/02/2024	1	3	144	113	48	383	27
	2	2	69	86	75	379	31
	3	3	122	107	117	449	19
19/03/2024	1	4	158	NA	NA	322	23
	2	2	91	53	60	289	2
	3	2	21	23	8	281	2
04/06/2024	1	7	85	117	135	390	22
	3	2	83	113	66	383	8
06/08/2024	1	8	207	153	150	362	19
	2	1	19	27	9	434	4
22/10/2024	1	5	250	161	149	356	6
	2	1	48	47	15	403	7
	3	1	58	53	19	328	7
Annual mean	1	5	169	136	120	363	19
	2	2	57	53	40	376	11
	3	2	71	74	52	360	9

3.1.4 Bioavailable nitrogen components

Nitrate (NO_3^-), nitrite (NO_2^-) and ammonium (NH_4^+) are forms of inorganic nitrogen that are readily accessible to organisms. Table 3.4 shows the concentrations of these three nitrogen (N) compounds in the sediment at the three sampling sites. Due to the limited amount of sediment material available, only one sample from Site 1 was analysed, therefore, mean values are only reported for Sites 2 and 3.

The highest individual concentration of nitrite + nitrate was measured at Site 1 (18.5 µg N g^{-1}). The concentrations of nitrite and nitrate showed a considerable variation across the sampling dates at both Site 2 and Site 3, with mean values of 4.86 µg N g^{-1} and 6.25 µg N g^{-1}, respectively. The mean concentrations of ammonium in the

sediment were 18.4 µg N g^{-1} at Site 2 and 18.8 µg N g^{-1} at Site 3, however, the ammonium concentrations also varied considerably across the samples, with values below the limit of detection (LOD) recorded in February and August 2024 at Site 2. Ammonium at Site 1 was below LOD, however, as shown in the results from the other two sites, the concentrations of this N compound can vary across the year and therefore, more samples are required to inform on the ammonium content in sediment at this site.

Table 3.4 Concentrations of available nitrogen in the sediment.

Date	Site	NO_2^- + NO_3^- (µg N g^{-1})	NH_4^+ (µg N g^{-1})
20/02/2024	1	18.5	<LOD
	2	13.3	<LOD
	3	11.8	29.5
19/03/2024	2	7.12	8.77
04/06/2024	2	1.40	49.8
06/08/2024	2	0.79	<LOD
22/10/2024	2	0.71	8.03
	3	1.71	33.5
Mean concentrations	2	4.86	18.4
	3	6.25	18.8

3.2 Sediment physical properties

3.2.1 Particle size distribution

The sediment grain size influences the amount of P in sediment, with the P content decreasing with increasing grain size (Kozyrev et al., 2023), increasing the potential for the P release from sediment into water. The particle size distribution and the particle size categories for the sediment at the three river sites are shown in the Table 3.5. Figure 3.4, Figure 3.5 and Figure 3.6 present each sediment fraction as percentage of total sediment at Sites 1, 2 and 3, respectively.

The annual average median grain sizes at Sites 1 and 3 were 18 µm and 55 µm, respectively, falling into the silt range category (i.e. between 4 µm and <63 µm). The sediment at Site 2 was, in general, coarser compared to the other two sites, with the average median grain size of 112 µm being in the sand range (i.e. ≥ 63 µm and < 1

mm). The sediment at Sites 1 and 3 was dominated by silt-size particles, which on average accounted for 76% and 56% of the total sediment, respectively, followed by sand-size material, the mean percentage of which was 14% at Site 1 and 38% at Site 3. The mean percentage of the clay-sized material (<4 µm) was 10% at Site 1 and 6% at Site 3. Sand was the dominant size fraction at Site 2, with the annual mean volume of 52% of the total sediment. The volume of finer sediment particles, clay and silt, was lower compared to the other two sites, accounting on average for 5% and 43%, respectively. Gravel was only measured at Site 2 in August, accounting for 0.1% of the total sediment.

Table 3.5 Selected percentiles for measures of sediment particle size distribution and percentage of clay, silt, sand and gravel volume in sediment at the three River Breamish sites in 2024.

Date	Site	Grain size (µm) at selected %iles			Clay % by volume	Silt % by volume	Sand % by volume	Gravel % by volume
		10%	50%	90%				
20/02/2024	1	4.51	20.8	82.1	7.66	78.1	14.2	0.00
	2	5.02	30.9	156	7.07	64.6	28.3	0.00
	3	4.19	22.8	90.0	9.34	73.8	16.9	0.00
19/03/2024	1	3.69	16.7	79.8	11.8	75.2	13.0	0.00
	2	9.72	122	335	2.47	33.4	64.2	0.00
04/06/2024	1	3.80	18.8	92.3	11.1	72.6	16.3	0.00
	2	5.45	40.0	202	6.22	57.0	36.8	0.00
	3	5.37	40.4	211	6.22	57.1	36.7	0.00
06/08/2024	1	4.40	18.3	98.3	8.21	74.4	17.4	0.00
	2	68.7	324	779	1.45	8.42	90.0	0.10
22/10/2024	1	3.88	15.4	70.1	10.8	78.4	10.7	0.00
	2	5.83	45.3	356	5.42	52.8	41.8	0.00
	3	8.39	101	463	3.77	36.2	60.0	0.00
Annual mean	1	4.1	18.0	84.5	9.91	75.7	14.3	0.00
	2	19.0	112	366	4.53	43.2	52.2	0.02
	3	6.0	54.7	255	6.44	55.7	37.9	0.00

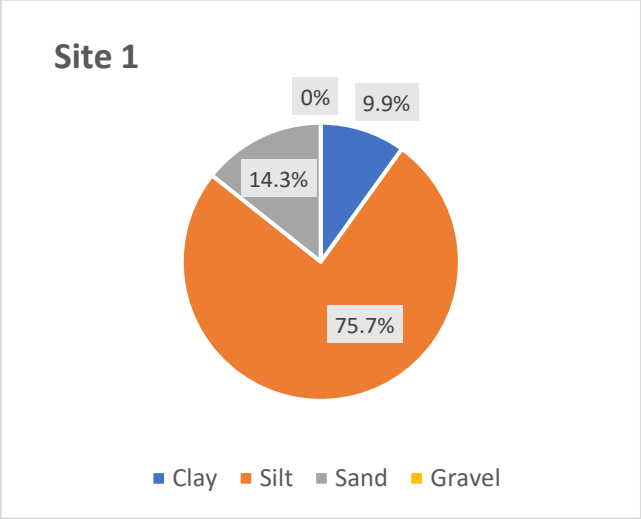


Figure 3.4 Percentage of clay, silt, sand and gravel volume in the sediment at Site 1.

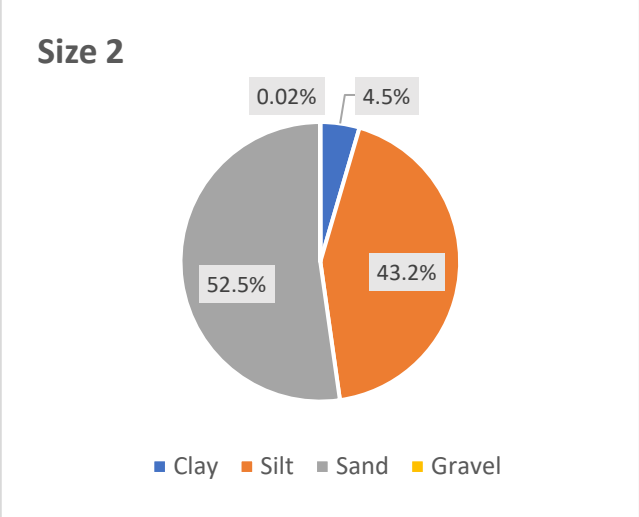


Figure 3.5 Percentage of clay, silt, sand and gravel volume in the sediment at Site 2.

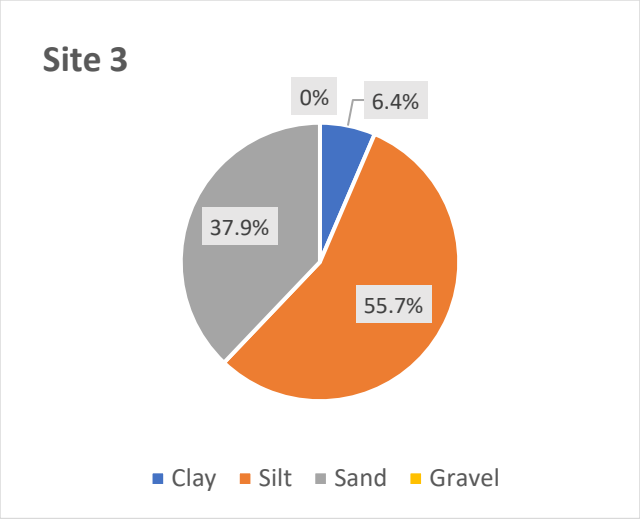


Figure 3.6 Percentage of clay, silt, sand and gravel volume in the sediment at Site 3.

Future interpretation of data

The data presented in this report provide baseline information on the chemical and physical properties of the sediment in the River Breamish prior to the implementation of the restoration measures. Although the number of sampling occasions was insufficient for a seasonal variation analysis, the data will allow the assessment of the indicator responses at an annual time scale. The Before-After-Control-Impact design of the monitoring will allow the assessment of natural variability within the river channel, including spatial variation.

4. References

Hupfer, M., R. Gächter, and Giovanoli, R. 1995. Transformation of phosphorus species in settling seston and during early sediment diagenesis. Aquatic Sciences 57, 305-324.

Hupfer, M., D. Zak, R. Roßberg, C. Herzog, and Pöthig, R. 2009. Evaluation of a well-established sequential phosphorus extraction technique for use in calcite-rich lake sediments: identification and prevention of artifacts due to apatite formation. Limnology & Oceanography: Methods 7, 399-410.

Kozyrev, R., Umezawa, Y. and Yoh, M. 2023.Total phosphorus and phosphorus forms change in sediments along the Tone River. Frontiers in Earth Science 11:1060312. doi.org/10.3389/feart.2023.1060312

Lewandowski, J., I. Schauser, and Hupfer, M. 2003. Long term effects of phosphorus precipitations with alum in hypereutrophic Lake Süsser See (Germany). Water Research 37, 3194-3204.

Meis, S., B.M. Spears, S. Maberly, and Perkins, R. 2012. Sediment amendment with Phoslock® in Clatto Reservoir (Dundee,UK): investigating spatial distribution & impacts on sediment phosphorus binding capacity. Journal of Environmental Management 93, 185-193.

Murphy, J. and Riley, J.P. 1962. A modified single solution method for the determination of phosphate in natural waters. Analytica Chemica Acta 27, 31-36.

Naden, P.S., Murphy, J.F., Old, G.H., Newman, J., Scarlett, P., Harman, M., Duerdoth, C.P., Hawczak, A., Pretty, J.L., Arnold, A., Laizé, C., Hornby, D.D., Collins, A.L., Sear, D.A. and Jones, J.I. 2016. Understanding the controls on deposited fine sediment in the streams of agricultural catchments. Science of The Total Environment 547, 366-381, https://doi.org/10.1016/j.scitotenv.2015.12.079.

Psenner, R., B. Boström, M. Dinka, K. Petterson, R. Pucsko, and M. Sager. 1988. Fractionation of phosphorus in suspended matter and sediment. Archive für Hydrobiologie: Ergebnisse der Limnologie 30, 98-103.

Spears, B.M., L. Carvalho, R. Perkins, A. Kirika, and D.M. Paterson. 2007. Sediment P cycling in a large shallow lake: spatio-temporal variation in P pools and release. Hydrobiologia, 584, 37-48.

Contact

enquiries@ceh.ac.uk

@UK_CEH

ceh.ac.uk



Bangor

UK Centre for Ecology & Hydrology
Environment Centre Wales
Deiniol Road
Bangor
Gwynedd
LL57 2UW

+44 (0)1248 374500

Edinburgh

UK Centre for Ecology & Hydrology
Bush Estate
Penicuik
Midlothian
EH26 0QB

+44 (0)131 4454343

Lancaster

UK Centre for Ecology & Hydrology
Lancaster Environment Centre
Library Avenue
Bailrigg
Lancaster
LA1 4AP

+44 (0)1524 595800

Wallingford (Headquarters)

UK Centre for Ecology & Hydrology
Maclean Building
Benson Lane
Crowmarsh Gifford
Wallingford
Oxfordshire
OX10 8BB

+44 (0)1491 838800

Disclaimer goes here ….

UK Centre for Ecology & Hydrology